

MediRemind - Detailed Architecture Document

AI Powered Medication Reminder & Health Assistant

1. Executive Summary

MediRemind is a secure healthcare application focused on medication adherence, prescription analysis, report summarization, and AI-assisted health support aligned with SDG 3.

2. Project Goals

Improve medication consistency, simplify health record access, provide AI-assisted summaries, and deliver timely reminders.

3. Stakeholders

Patients, caregivers (future), healthcare professionals (future), administrators.

4. Functional Requirements

Authentication, profile management, medication management, reminders, report uploads, AI summaries, AI assistant, dashboard analytics.

5. Non-Functional Requirements

Security, scalability, maintainability, availability, responsiveness, usability.

6. Technology Stack

Frontend: Kotlin + Jetpack Compose. Backend: FastAPI. Database: MongoDB Atlas. AI: Gemini 2.5 Flash. Notifications: FCM. Storage: Cloudinary.

7. High Level Architecture

Android App -> FastAPI -> MongoDB Atlas / Cloudinary / Gemini API / Firebase Cloud Messaging.

8. Android Architecture

Clean Architecture + MVVM. Layers: UI, ViewModel, Domain, Repository, Data, Network.

9. Backend Architecture

Routes -> Services -> Repositories -> Database. Business logic isolated from routes.

10. Database Design

Collections: users, medications, reports, chat_history. User-centric data ownership model.

11. Authentication Flow

Register -> Hash Password -> Login -> JWT Access Token -> Authorized Requests.

12. Medication Workflow

Add Medication -> Schedule Reminder -> Notification -> Taken/Skip/Remind Later -> Update Consistency Score.

13. Prescription Extraction Workflow

Upload File -> Cloudinary -> Gemini Extraction -> Review Medicines -> Approve/Edit -> Save.

14. Report Summarization Workflow

Upload Report -> Gemini Summary -> Store Summary -> Available to AI Assistant.

15. AI Assistant Architecture

Context Builder uses profile, medications, and report summaries. Raw reports are not repeatedly sent.

16. AI Safety Framework

No diagnosis, no prescribing, no medication discontinuation advice, supportive tone, healthcare disclaimer.

17. Dashboard Design

Consistency Ring, Progress Ring, Donut Chart, Timeline, AI Insights, Reports, Statistics Cards.

18. Security Architecture

JWT, bcrypt, HTTPS, validation, file restrictions, user-level authorization.

19. Scalability Plan

Stateless APIs, modular services, cloud storage, scalable database, future caching support.

20. API Modules

Auth APIs, User APIs, Medication APIs, Reports APIs, Dashboard APIs, AI APIs, Reminder APIs.

21. Deployment Architecture

Android App (Play Store), FastAPI on cloud host, MongoDB Atlas, Cloudinary, Firebase, Gemini.

22. Testing Strategy

Unit tests, integration tests, API tests, UI tests, validation testing, security testing.

23. Future Scope

Doctor Portal, Caregiver Access, Smartwatch Support, Drug Interaction Detection, Offline Mode.

24. Development Roadmap

Phase 1 Architecture, Phase 2 Backend, Phase 3 Frontend, Phase 4 AI, Phase 5 Testing, Phase 6 Deployment.